



香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

General Chemistry CHM1001

Lecture 3

Metals and nonmetals (Chapter 7)

Basic Concepts of Chemical Bonding (Chapter 8)

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About Exams

What forms will we choose--open-book examination or others ?

=> Closed-book exam, but you may bring one cheat sheet (double-sided allowed)

What content would the exam cover ?

=> All material covered in class.

Does it necessary for us to memorize the definition and description of the terms?

=> Yes

Is it similar to the practices in our textbook ?

=> Yes, and it is also similar to the assignments.

Midterm and Final Exams

Electronic Structure

Periodic properties

Chemical Bonding

Molecular Geometries

Gases

Liquids and IM forces

Solids and Materials

Chemical Equilibrium

Solutions

Acid-Base Equilibrium

Additional Asp soln

Chemical Kinetics

Chemical Thermodynamics

Thermochemistry

Environment

Organic and Biological

Electrochemistry

Nuclear Chemistry

Periodic Table

1A																7A	8A
H ⁺																H ⁻	N O B L E G A S E S
2A												3A	4A	5A	6A		
Li ⁺														N ³⁻	O ²⁻	F ⁻	
Na ⁺	Mg ²⁺	Transition metals										Al ³⁺		P ³⁻	S ²⁻	Cl ⁻	
K ⁺	Ca ²⁺	Sc ³⁺	Ti ⁴⁺	V ⁵⁺ V ⁴⁺	Cr ³⁺	Mn ²⁺ Mn ⁴⁺	Fe ²⁺ Fe ³⁺	Co ²⁺ Co ³⁺	Ni ²⁺	Cu ⁺ Cu ²⁺	Zn ²⁺				Se ²⁻	Br ⁻	
Rb ⁺	Sr ²⁺								Pd ²⁺	Ag ⁺	Cd ²⁺		Sn ²⁺ Sn ⁴⁺	Sb ³⁺ Sb ⁵⁺	Te ²⁻	I ⁻	
Cs ⁺	Ba ²⁺								Pt ²⁺	Au ⁺ Au ³⁺	Hg ₂ ²⁺ Hg ²⁺		Pb ²⁺ Pb ⁴⁺	Bi ³⁺ Bi ⁵⁺			

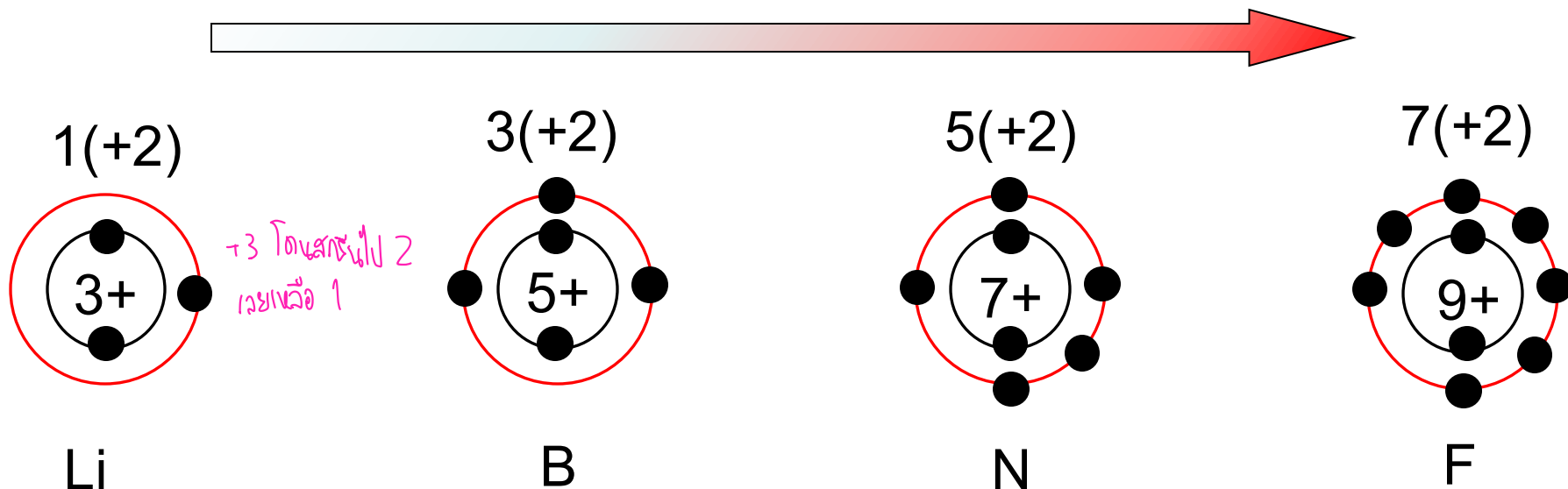
Periodic Table

110080 1 H Hydrogen Nonmetal																		4.00260 2 He Helium Noble Gas																																																																																																																																																																																																																																																																																																																	
7.0 3 Li Lithium Alkali Metal																		9.012183 4 Be Beryllium Alkaline Earth Metal																		26.981538 5 B Boron Metalloid																		12.011 6 C Carbon Nonmetal																		14.007 7 N Nitrogen Nonmetal																		15.999 8 O Oxygen Nonmetal																		18.9984 9 F Fluorine Halogen																		20.180 10 Ne Neon Noble Gas																																																																																																																																																																																																					
22.98976928 11 Na Sodium Alkali Metal																		24.304 12 Mg Magnesium Alkaline Earth Metal																		26.981538 13 Al Aluminum Post-Transition Metal																		28.0855 14 Si Silicon Metalloid																		30.973761 15 P Phosphorus Nonmetal																		32.07 16 S Sulfur Nonmetal																		35.45 17 Cl Chlorine Halogen																		39.9 18 Ar Argon Noble Gas																																																																																																																																																																																																					
39.0983 19 K Potassium Alkali Metal																		40.08 20 Ca Calcium Alkaline Earth Metal																		44.95591 21 Sc Scandium Transition Metal																		47.867 22 Ti Titanium Transition Metal																		50.9415 23 V Vanadium Transition Metal																		51.996 24 Cr Chromium Transition Metal																		54.93804 25 Mn Manganese Transition Metal																		55.84 26 Fe Iron Transition Metal																		58.93319 27 Co Cobalt Transition Metal																		58.693 28 Ni Nickel Transition Metal																		63.55 29 Cu Copper Transition Metal																		65.4 30 Zn Zinc Transition Metal																		69.723 31 Ga Gallium Post-Transition Metal																		72.63 32 Ge Germanium Metalloid																		74.92159 33 As Arsenic Metalloid																		78.97 34 Se Selenium Nonmetal																		79.90 35 Br Bromine Halogen																		83.80 36 Kr Krypton Noble Gas																	
85.468 37 Rb Rubidium Alkali Metal																		87.62 38 Sr Strontium Alkaline Earth Metal																		88.90584 39 Y Yttrium Transition Metal																		91.22 40 Zr Zirconium Transition Metal																		92.90637 41 Nb Niobium Transition Metal																		95.95 42 Mo Molybdenum Transition Metal																		96.90636 43 Tc Technetium Transition Metal																		101.1 44 Ru Ruthenium Transition Metal																		102.9055 45 Rh Rhodium Transition Metal																		106.42 46 Pd Palladium Transition Metal																		107.868 47 Ag Silver Transition Metal																		112.41 48 Cd Cadmium Transition Metal																		114.818 49 In Indium Post-Transition Metal																		118.71 50 Sn Tin Post-Transition Metal																		121.760 51 Sb Antimony Metalloid																		127.6 52 Te Tellurium Metalloid																		126.9045 53 I Iodine Halogen																		131.29 54 Xe Xenon Noble Gas																	
132.905 55 Cs Cesium Alkali Metal																		137.33 56 Ba Barium Alkaline Earth Metal																		178.49 72 Hf Hafnium Transition Metal																		180.9479 73 Ta Tantalum Transition Metal																		183.84 74 W Tungsten Transition Metal																		186.207 75 Re Rhenium Transition Metal																		190.2 76 Os Osmium Transition Metal																		192.22 77 Ir Iridium Transition Metal																		195.08 78 Pt Platinum Transition Metal																		196.967 79 Au Gold Transition Metal																		200.59 80 Hg Mercury Transition Metal																		204.383 81 Tl Thallium Post-Transition Metal																		207 82 Pb Lead Post-Transition Metal																		208.980 83 Bi Bismuth Post-Transition Metal																		208.980 84 Po Polonium Metalloid																		209.987 85 At Astatine Halogen																		222.017 86 Rn Radon Noble Gas																																			
223.019 87 Fr Francium Alkali Metal																		226.025 88 Ra Radium Alkaline Earth Metal																		267.101 104 Rf Rutherfordium Transition Metal																		268.101 105 Db Dubnium Transition Metal																		269.101 106 Sg Seaborgium Transition Metal																		270.101 107 Bh Bohrium Transition Metal																		269.101 108 Hs Hassium Transition Metal																		277.101 109 Mt Meitnerium Transition Metal																		282.101 110 Ds Darmstadtium Transition Metal																		282.101 111 Rg Roentgenium Transition Metal																		286.101 112 Cn Copernicium Transition Metal																		286.101 113 Nh Nihonium Post-Transition Metal																		290.101 114 Fl Flerovium Post-Transition Metal																		290.101 115 Mc Moscovium Post-Transition Metal																		293.201 116 Lv Livermorium Post-Transition Metal																		294.201 117 Ts Tennesine Halogen																		295.201 118 Og Oganesson Noble Gas																																			
138.905 57 La Lanthanum Lanthanide																		140.116 58 Ce Cerium Lanthanide																		140.908 59 Pr Praseodymium Lanthanide																		144.24 60 Nd Neodymium Lanthanide																		144.913 61 Pm Promethium Lanthanide																		150.4 62 Sm Samarium Lanthanide																		151.964 63 Eu Europium Lanthanide																		157.2 64 Gd Gadolinium Lanthanide																		158.925 65 Tb Terbium Lanthanide																		162.500 66 Dy Dysprosium Lanthanide																		164.930 67 Ho Holmium Lanthanide																		167.26 68 Er Erbium Lanthanide																		168.935 69 Tm Thulium Lanthanide																		173.05 70 Yb Ytterbium Lanthanide																		174.967 71 Lu Lutetium Lanthanide																																																																							
227.028 89 Ac Actinium Actinide																		232.038 90 Th Thorium Actinide																		231.036 91 Pa Protactinium Actinide																		238.029 92 U Uranium Actinide																		237.043 93 Np Neptunium Actinide																		244.064 94 Pu Plutonium Actinide																		243.061 95 Am Americium Actinide																		247.063 96 Cm Curium Actinide																		247.063 97 Bk Berkelium Actinide																		251.063 98 Cf Californium Actinide																		252.083 99 Es Einsteinium Actinide																		257.083 100 Fm Fermium Actinide																		258.103 101 Md Mendelevium Actinide																		259.103 102 No Nobelium Actinide																		266.103 103 Lr Lawrencium Actinide																																																																							

Recap

Effective Nuclear Charge (ENC)

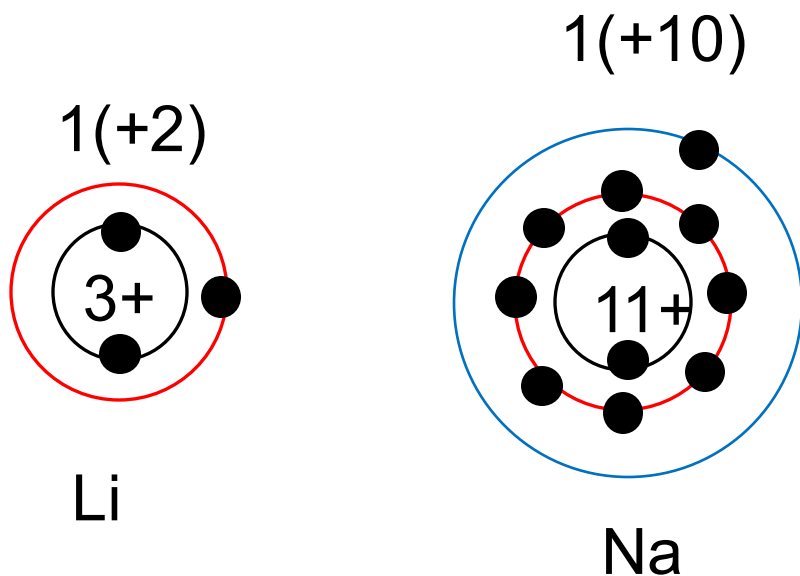
The effective nuclear charge (ENC) is the **actual amount of positive (nuclear) charge experienced by an electron** in a multi-electron atom.



Recap

Effective Nuclear Charge (ENC)

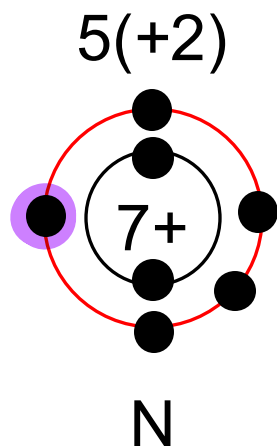
The effective nuclear charge (ENC) is the actual amount of positive (nuclear) charge experienced by an electron in a multi-electron atom.



Note: this explanation is a simplification; in reality, the shielding effect is different for different s orbital p orbitals. To obtain a more accurate value, use the Slater equation, etc.

Slater's Rule

- **Electrons in the same group** as the target electron: 0.35 each
- **Electrons in the next inner shell**: 0.85 each
- **Electrons in all deeper shells**: 1.00 each



$$Z_{\text{eff}} = Z - S \quad (1s^2)(2s^2 2p^3)$$

5 valence $e^- \rightarrow$ ดูแรงที่อีก 4 ตัวที่เปลือกนอก

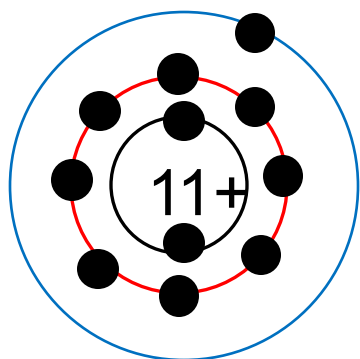
- Same group ($2s^2 2p^3$, excluding the target electron): 4 electrons $\times 0.35 = \mathbf{1.40}$
- One shell lower ($1s^2$): 2 electrons $\times 0.85 = \mathbf{1.70}$
- Total $S = 1.40 + 1.70 = 3.10$

$$Z_{\text{eff}} = 7 - 3.10 = \mathbf{3.90} \quad (\text{คำนวณเปลือก} = 5)$$

Slater's Rule

- **Electrons in the same group** as the target electron: 0.35 each
- **Electrons in the next inner shell**: 0.85 each
- **Electrons in all deeper shells**: 1.00 each

1(+10)



$$Z_{\text{eff}} = Z - S \quad (1s^2)(2s^2 2p^6)(3s^1)$$

- Same group: 0 electrons $\times 0.35 = 0.00$
- One shell lower (2s² 2p⁶): 8 electrons $\times 0.85 = 6.80$
- Deeper shells (1s²): 2 electrons $\times 1.00 = \mathbf{2.00}$

Na

$$\text{Total } S = 0.00 + 6.80 + 2.00 = 8.80$$

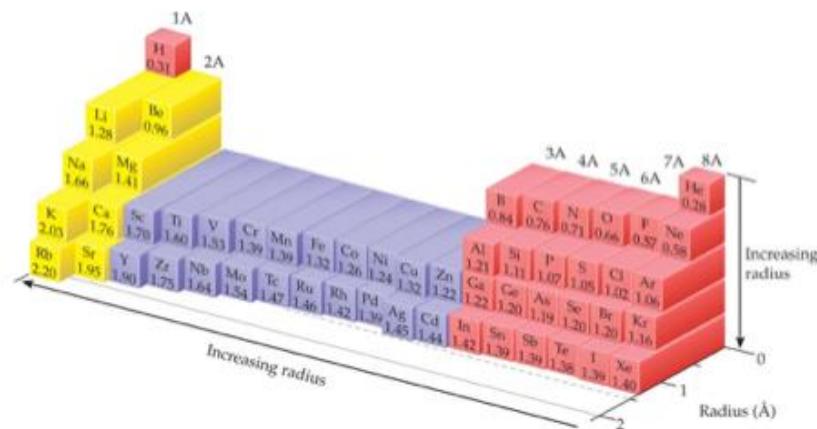
$$Z_{\text{eff}} = 11 - 8.80 = \mathbf{2.20} \quad (\text{ค่าคงที่} = 1)$$

Recap

Sizes of Atoms

#p⁺ ↑, #e⁻ ↑ → more strongly attracted to nucleus

- The bonding atomic radius is half the internuclear distance when atoms are bonded.
- The bonding atomic radius tends to
 - decrease from left to right across a period (Z_{eff} ↑).
 - increase from top to bottom of a group (n ↑).



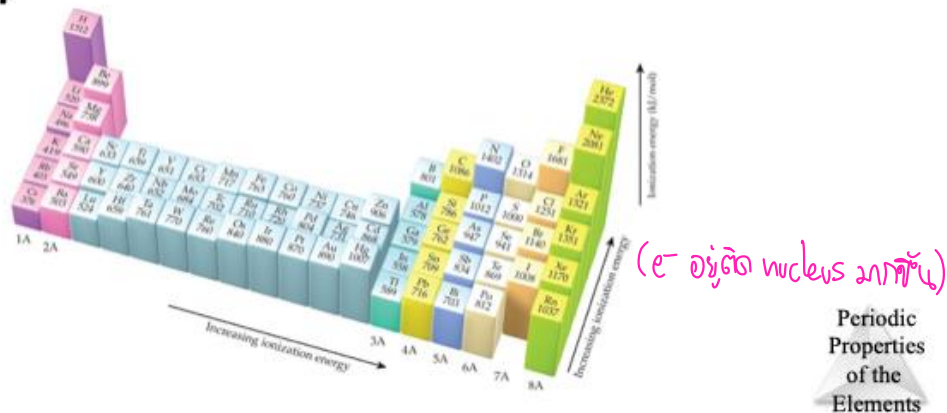
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Periodic
Properties
of the
Elements

Recap

Factors that Influence Ionization Energy

- Smaller atoms have higher ^{IE} / values.
- / values depend on effective nuclear charge and average distance of the electron from the nucleus.



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Recap

General Trend in Electron Affinity

- Not much change in a group.
- Across a period, it generally increases. *Three* notable exceptions include the following:

- 1) Group 2A: *s* sublevel is full!
- 2) Group 5A: *p* sublevel is half-full!

- 3) Group 8A: *p* sublevel is full!

Note: the electron affinity for many of these elements is *positive* (X^- is unstable).

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1A	2A	3A	4A	5A	6A	7A	8A
H -73	Be > 0	B -27	C -122	N > 0	O -141	F -328	He > 0
Li -60	Mg > 0	Al -43	Si -134	P -72	S -200	Cl -349	Ne > 0
Na -53	K -48	Ga -30	Ge -119	As -78	Se -195	Br -325	Ar > 0
Rb -47	Sr -5	In -30	Sn -107	Sb -103	Te -190	I -295	Kr > 0
						Xe > 0	

s sublevel | เต็ม
แล้ว e- ใหม่อีก

ถ้าเต็มแล้ว 1s=2 → รับ e- ใหม่ไม่ได้

Periodic
Properties
of the
Elements

Recap

Irregularities in the General Trend

- The trend is **not** followed when the added valence electron in the next element
 - enters a new sublevel (higher energy sublevel);
 - is the first electron to pair in one orbital of the sublevel (electron repulsions lower energy).



Oxygen



Nitrogen

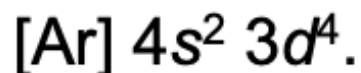
Recap

Chromium as an Anomaly

- For instance, the electron configuration for chromium is



rather than the expected



- This occurs because the 4s and 3d orbitals are very close in energy.
- These anomalies occur in *f*-block atoms with *f* and *d* orbitals, as well.

Metals Differ from Nonmetals

- Metals tend to form cations.
- Nonmetals tend to form anions.

1A																7A	8A
H ⁺																H ⁻	N O B L E G A S E S
Li ⁺														N ³⁻	O ²⁻	F ⁻	
Na ⁺	Mg ²⁺	Transition metals										Al ³⁺		P ³⁻	S ²⁻	Cl ⁻	
K ⁺	Ca ²⁺	Sc ³⁺	Ti ⁴⁺	V ⁵⁺ V ⁴⁺	Cr ³⁺	Mn ²⁺ Mn ⁴⁺	Fe ²⁺ Fe ³⁺	Co ²⁺ Co ³⁺	Ni ²⁺	Cu ⁺ Cu ²⁺	Zn ²⁺				Se ²⁻	Br ⁻	
Rb ⁺	Sr ²⁺								Pd ²⁺	Ag ⁺	Cd ²⁺		Sn ²⁺ Sn ⁴⁺	Sb ³⁺ Sb ⁵⁺	Te ²⁻	I ⁻	
Cs ⁺	Ba ²⁺								Pt ²⁺	Au ⁺ Au ³⁺	Hg ₂ ²⁺ Hg ²⁺		Pb ²⁺ Pb ⁴⁺	Bi ³⁺ Bi ⁵⁺			

Increasing metallic character																	
1A 1																	8A 18
1 H	2A 2											3A 13	4A 14	5A 15	6A 16	7A 17	2 He
3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
11 Na	12 Mg	3B 3	4B 4	5B 5	6B 6	7B 7	8B 8 9 10			1B 11	2B 12	13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
55 Cs	56 Ba	71 Lu	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
87 Fr	88 Ra	103 Lr	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cp	113	114 Fl	115	116 Lv	117	118

Increasing metallic character

Metals

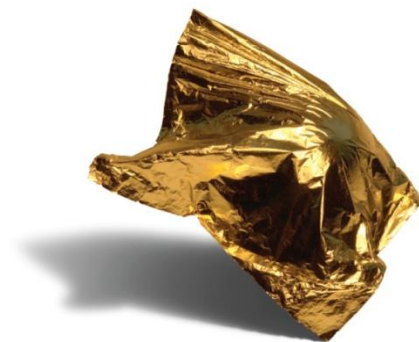
Metalloids

Nonmetals

57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb
89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No

Metals

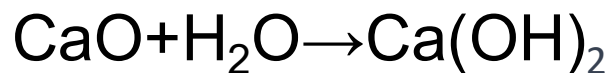
- ❖ Most of the elements in nature are metals.
- ❖ Properties of metals:
 - Shiny luster
 - Conduct heat and electricity
 - Malleable and ductile
 - Solids at room temperature (except mercury)
 - Low ionization energies/form cations easily



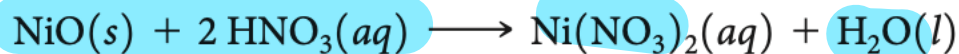
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Metal Chemistry

- Compounds formed between **metals** and **nonmetals** tend to be **ionic**.
anion *cation*
- Metal oxides** tend to be **basic**.



Metal oxide + acid $\longrightarrow$ salt + water

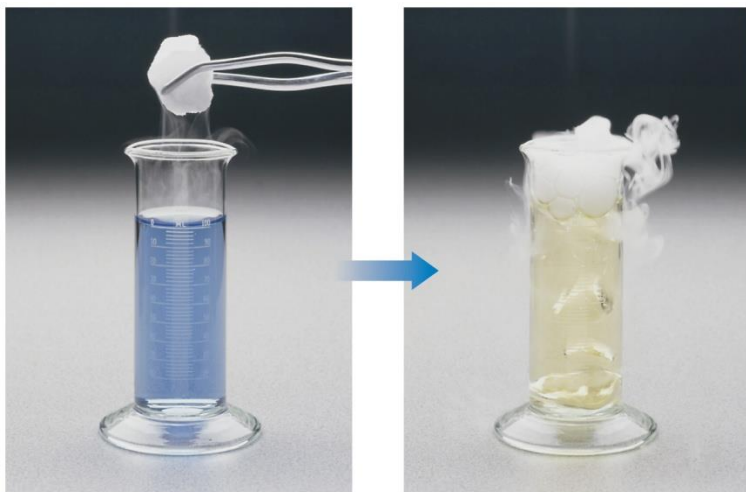


Nonmetals

- Nonmetals are found on the right hand side of the periodic table.
- Properties of nonmetals include the following:
 - Solid, liquid, or gas (depends on element)
 - Solids are dull, brittle, poor conductors
 - Large negative electronegativity/form anions readily



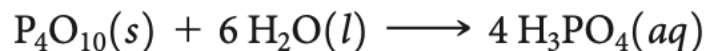
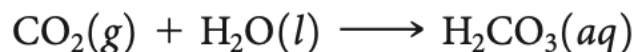
Nonmetal Chemistry



The reaction of CO_2 with water containing a bromthymol blue indicator. Initially, the blue color tells us the water is slightly basic. When a piece of solid carbon dioxide (“dry ice”) is added, the color changes to yellow, indicating an acidic solution. The mist is water droplets condensed from the air by the cold CO_2 gas.

- Substances containing only nonmetals are molecular compounds.
- Most nonmetal oxides are acidic.

Nonmetal oxide + water $\longrightarrow$ acid



Properties of Metals and Nonmetals

Table 7.3 Characteristic Properties of Metals and Nonmetals

Metals	Nonmetals
Have a shiny luster; various colors, although most are silvery	Do not have a luster; various colors
Solids are malleable and ductile	Solids are usually brittle; some are hard, and some are soft
Good conductors of heat and electricity	Poor conductors of heat and electricity
Most metal oxides are ionic solids that are basic	Most nonmetal oxides are molecular substances that form acidic solutions
Tend to form cations in aqueous solution	Tend to form anions or oxyanions in aqueous solution

Metalloids

- Metalloids have some characteristics of metals and some of nonmetals.
- Several metalloids, such as Si, are electrical semiconductors (computer chips).



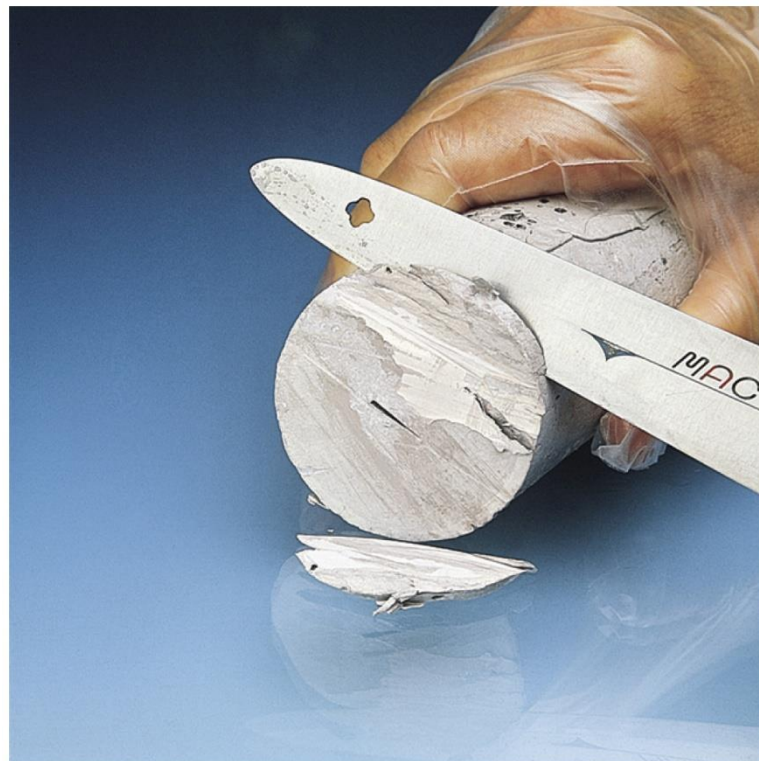
Group Trends

- Elements in a group have similar properties.
- Trends also exist within groups.
- Groups Compared:
 - Group 1A: The Alkali Metals
 - Group 2A: The Alkaline Earth Metals
 - Group 6A: The Oxygen Group
 - Group 7A: The Halogens
 - Group 8A: The Noble Gases

Alkali Metals

မာရ်သီယံ

- **Alkali metals** are **soft, metallic solids**.
- They are found only in compounds in nature, not in their elemental forms.
- **Typical metallic properties** (luster, conductivity) are seen in them.



Alkali Metal Properties

- They have low densities and melting points.
- They also have low ionization energies.

Table 7.4 Some Properties of the Alkali Metals

Element	Electron Configuration	Melting Point (°C)	Density (g/cm ³)	Atomic Radius (Å)	<i>I</i> ₁ (kJ/mol)
Lithium	[He]2s ¹	181	0.53	1.28	520
Sodium	[Ne]3s ¹	98	0.97	1.66	496
Potassium	[Ar]4s ¹	63	0.86	2.03	419
Rubidium	[Kr]5s ¹	39	1.53	2.20	403
Cesium	[Xe]6s ¹	28	1.88	2.44	376

Alkali Metal Chemistry



Li



Na

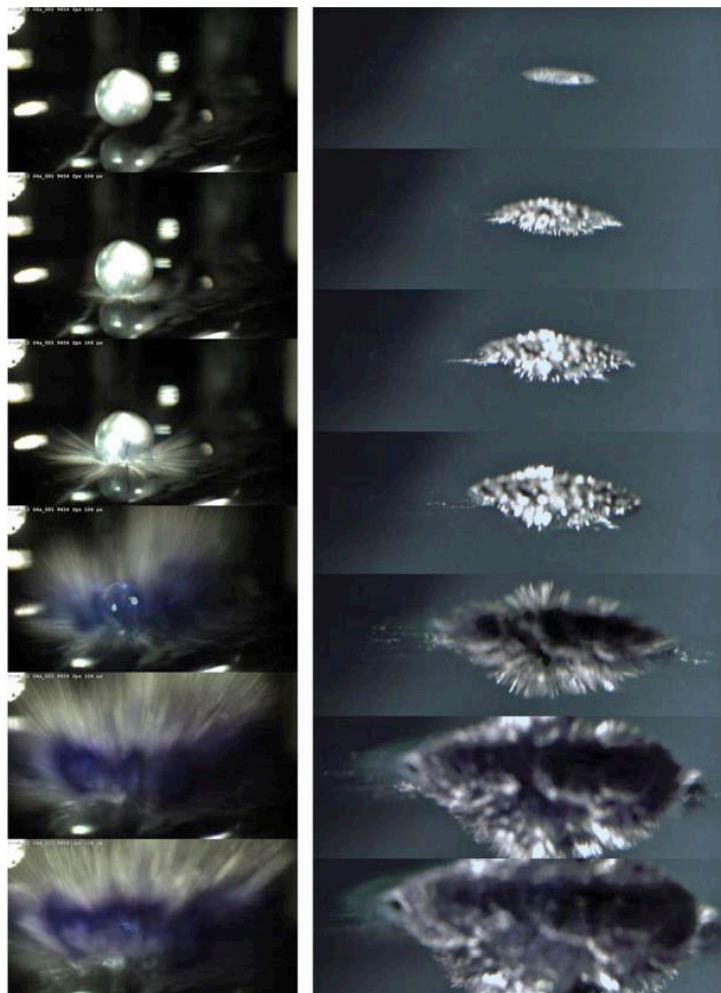


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Their reactions with water are famously exothermic.

[Nature Chemistry](#) volume 7, pages 250–254 (2015)

Alkali Metal Chemistry



Na ง่าย e- ให้โมลกุลน้ำ

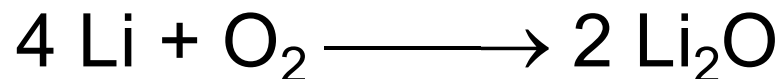
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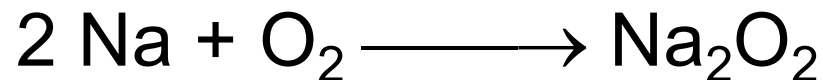
Nature Chemistry volume 7, pages 250–254 (2015)

Differences in Alkali Metal Chemistry

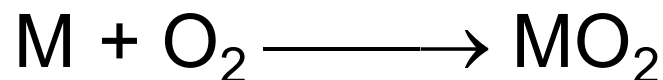
- Lithium reacts with oxygen to make an oxide:



- Sodium reacts with oxygen to form a peroxide:

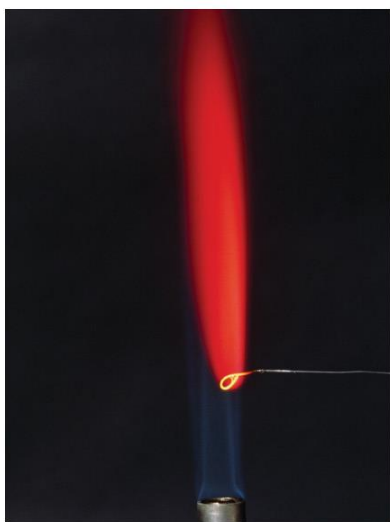


- K, Rb, and Cs also form superoxides:

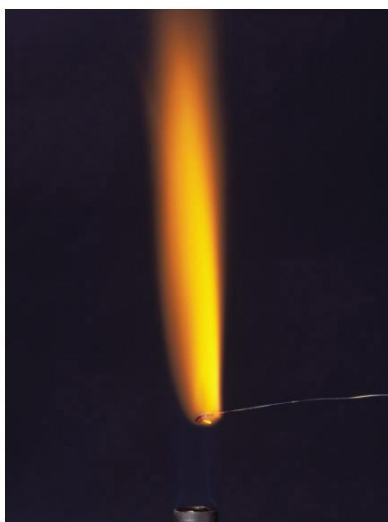


Flame Tests

- Qualitative tests for alkali metals include their characteristic colors in flames.



Li



Na



K

The Hydrogen Spectrum



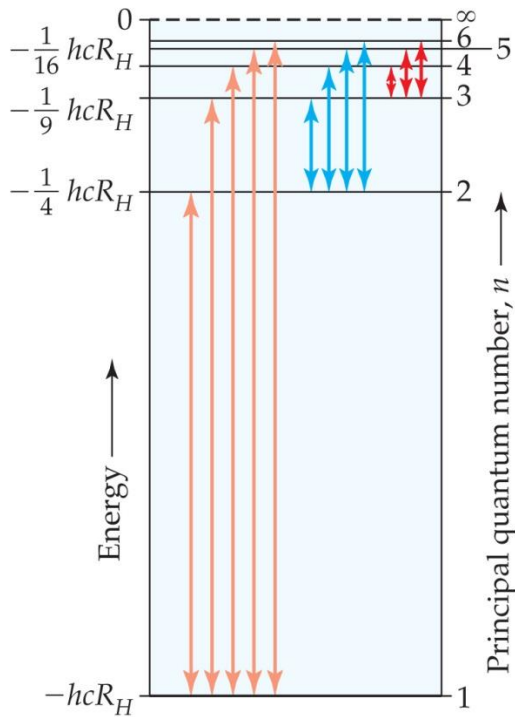
- Johann Balmer (1885) discovered a simple formula relating the four lines to integers.
- Johannes Rydberg advanced this formula.

$$\frac{1}{\lambda} = (R_H) \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

เปลี่ยนพลังงานเป็นโฟตอน ปล่อย ground state

- Neils Bohr explained *why* this mathematical relationship works.

The Bohr Model



The energy absorbed or emitted from the process of electron promotion or demotion can be calculated by the equation

$$\Delta E = -hcR_H \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

where R_H is the Rydberg constant, $1.097 \times 10^7 \text{ m}^{-1}$, and n_i and n_f are the initial and final energy levels of the electron.

Alkaline Earth Metals—Compare to Alkali Metals

Table 7.5 Some Properties of the Alkaline Earth Metals

Element	Electron Configuration	Melting Point (°C)	Density (g/cm ³)	Atomic Radius (Å)	I ₁ (kJ/mol)
Beryllium	[He]2s ²	1287	1.85	0.96	899
Magnesium	[Ne]3s ²	650	1.74	1.41	738
Calcium	[Ar]4s ²	842	1.55	1.76	590
Strontium	[Kr]5s ²	777	2.63	1.95	549
Barium	[Xe]6s ²	727	3.51	2.15	503

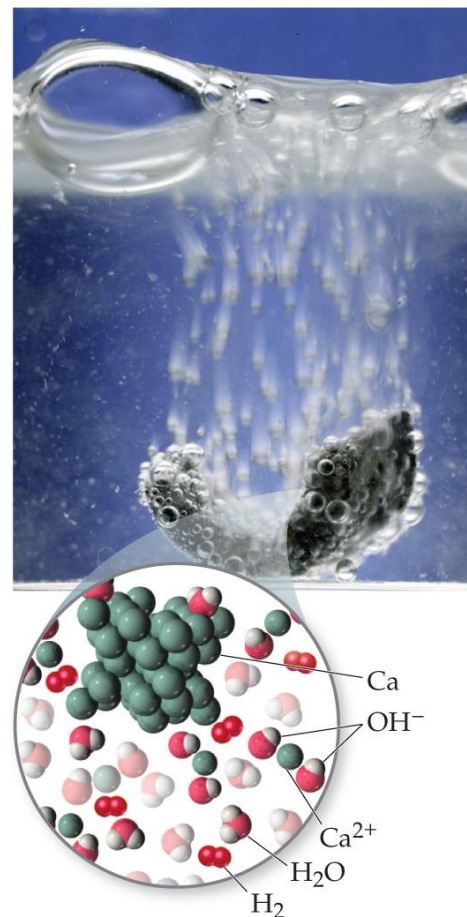
- Alkaline earth metals have higher densities and melting points than alkali metals.
- Their ionization energies are low, but not as low as those of alkali metals. *easily oxidized*

How important is GPA in your career?

I recently visited the U.S. East Coast and met with top universities including MIT. All of them wanted me to bring back one message: international students are a very important part of their community, and they hope we can recommend the very best students whenever possible. They also emphasized that GPA is not the most important factor. Instead, they place greater value on research experience and strong letters of recommendation from professors.

Alkaline Earth Metals

- Beryllium does not react with water, and magnesium reacts only with steam, but the other alkaline earth metals react readily with water.
- Reactivity tends to increase as you go down the group, due to the decrement of the ionization energies.



Group 6A—Increasing in Metallic Character down the Group

Table 7.6 Some Properties of the Group 6A Elements

Element	Electron Configuration	Melting Point (°C)	Density	Atomic Radius (Å)	I_1 (kJ/mol)
Oxygen	[He] $2s^22p^4$	-218	1.43 g/L	0.66	1314
Sulfur	[Ne] $3s^23p^4$	115	1.96 g/cm ³	1.05	1000
Selenium	[Ar] $3d^{10}4s^24p^4$	221	4.82 g/cm ³	1.20	941
Tellurium	[Kr] $4d^{10}5s^25p^4$	450	6.24 g/cm ³	1.38	869
Polonium	[Xe] $4f^{14}5d^{10}6s^26p^4$	254	9.20 g/cm ³	1.40	812

- Oxygen, sulfur, and selenium are nonmetals.
- Tellurium is a metalloid.
- The radioactive polonium is a metal.

(Down the group, there is an increase in atomic radius and a decrease in ionization energy)

Group 7A—Halogens

Table 7.7 Some Properties of the Halogens

Element	Electron Configuration	Melting Point (°C)	Density	Atomic Radius (Å)	I_1 (kJ/mol)
Fluorine	$[\text{He}]2s^22p^5$	-220	1.69 g/L	0.57	1681
Chlorine	$[\text{Ne}]3s^23p^5$	-102	3.12 g/L	1.02	1251
Bromine	$[\text{Ar}]4s^23d^{10}4p^5$	-7.3	3.12 g/cm ³	1.20	1140
Iodine	$[\text{Kr}]5s^24d^{10}5p^5$	114	4.94 g/cm ³	1.39	1008

- The **halogens** are typical nonmetals.
- They have highly negative electron affinities, so they exist as anions in nature.
- They react directly with metals to form metal halides.

Group 8A—Noble Gases

Table 7.8 Some Properties of the Noble Gases

Element	Electron Configuration	Boiling Point (K)	Density (g/L)	Atomic Radius* (Å)	I_1 (kJ/mol)
Helium	$1s^2$	4.2	0.18	0.28	2372
Neon	$[\text{He}]2s^22p^6$	27.1	0.90	0.58	2081
Argon	$[\text{Ne}]3s^23p^6$	87.3	1.78	1.06	1521
Krypton	$[\text{Ar}]4s^23d^{10}4p^6$	120	3.75	1.16	1351
Xenon	$[\text{Kr}]5s^24d^{10}5p^6$	165	5.90	1.40	1170
Radon	$[\text{Xe}]6s^24f^{14}5d^{10}6p^6$	211	9.73	1.50	1037

*Only the heaviest of the noble-gas elements form chemical compounds. Thus, the atomic radii for the lighter noble gas elements are estimated values.

- The noble gases have *stable / not prepared to lose e^-* very large ionization energies.
- Their electron affinities are positive (can't form stable anions).
- Therefore, they are relatively unreactive.
- They are found as monatomic gases.